IND-QUASI-AFFINENESS OF A COMPOSITION FORCES IND-QUASI-AFFINENESS OF THE FIRST FACTOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19620-19624 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19626-19632, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Permanence sibling of the collected composition law L16116; third card of the ind-quasi-affine series (dictionary, composition, this permanence).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter alternative-definition dictionary for ind-quasi-affine morphisms (card L16109); basic affineness permanence of quasi-affine morphisms over affine bases.

Source reference. Original lemma 0F1U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19538–19554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *If $f : X \to Y$ and $g : Y \to Z$ are morphisms of schemes such that $g \circ f$ is ind-quasi-affine, then f is ind-quasi-affine.*

Proof. By Lemma 0F1U, we may work Zariski locally on Z and then on Y, so we lose no generality by assuming that Z, and then also Y, is affine. Then any quasi-compact open of X is quasi-affine, so Lemma 0F1U gives the claim. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.